Small Molecule. Big Possibilities.

CITRIC ACID

The Chemistry of Sourness

A NATURAL MOLECULE. A BRIGHTER WORLD.

From Citrus Fruits to Everyday Products.

- FOOD
- HEALTH
- CLEANER PLANET
- BRIGHTER TOMORROW

1 COMPOUND IDENTITY

Common name: Citric Acid

IUPAC name:
2-hydroxypropane-1,2,3-tricarboxylic acid

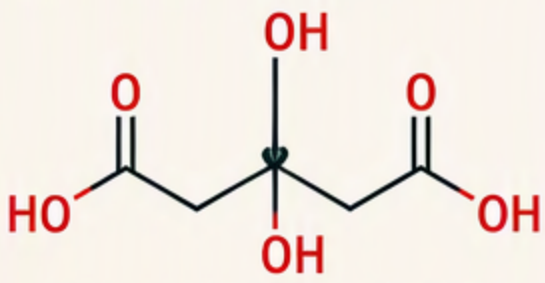
Molecular formula: $C_6H_8O_7$

Molar mass: 192.12 g/mol

CAS number: 77-92-9

A naturally occurring organic acid found in citrus fruits.

2 STRUCTURE



2-hydroxypropane-1,2,3-tricarboxylic acid
(citric acid)

Functional groups:

- Three carboxylic acid groups (–COOH)
- One hydroxyl group (–OH)
- Classified as a tricarboxylic acid

Three acids, one amazing molecule!

3 IMPORTANT PROPERTIES



Appearance:
White crystalline solid



Solubility:
Highly soluble in water (≈ 592 g/L at 25 °C)



Acidity:
Weak organic acid (pK_{a1} = 3.13 pK_{a2} = 4.76 pK_{a3} = 6.40)



Melting point:
153 °C (decomposes)

Its acidic nature gives citrus fruits their characteristic sour taste.

4 WHAT IS AMAZING ABOUT IT?

Citric acid naturally occurs in lemons, oranges, and other citrus fruits. It is one of the most widely used food additives in the world and can also be produced on a large scale through microbial fermentation.



"The molecule that gives life its zest."

5 USES AND VALUE



Food and beverage products
(acidity regulator, flavoring, preservative)



Pharmaceutical formulations
(e.g., effervescent tablets)



Cleaning products
(descaling agent, biodegradable cleaner)



Cosmetics
(pH adjustment, exfoliants)

A versatile compound that adds value across many industries.

6 RISKS AND SAFETY



- Generally low toxicity.
- May cause skin, eye, or respiratory irritation in concentrated form.
- Avoid inhalation of dust.
- Handle according to standard laboratory and industrial safety practices.







Safe when used properly, but always handle with care.

7 CHEMICAL ENGINEERING CONNECTION



Citric acid is commercially produced by fermentation (e.g., using *Aspergillus niger*), followed by filtration, purification, and crystallization. Chemical engineers design and optimize:

- Fermentation processes
- Downstream separation systems
- Crystallization and drying
- Waste treatment and sustainability strategies

Engineering turns simple molecules into global solutions.

8 REFERENCES

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